

An Empirical Theory of Asset Pricing

Vedant Sachdeva, October 19th, 2024

The Theoretical Model

$$dX = (\mu X)dt + \eta X dW$$

The Theoretical Model

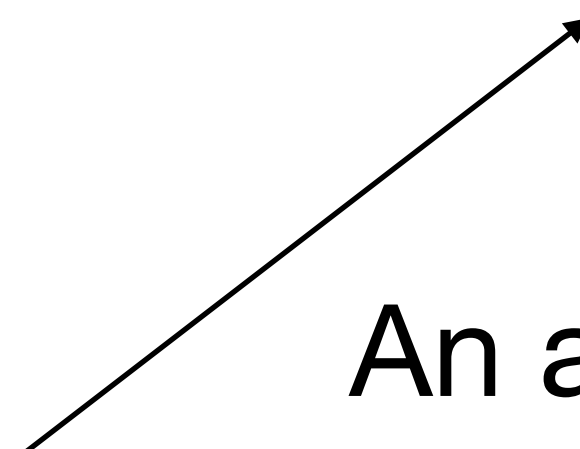
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An asset price

The Theoretical Model

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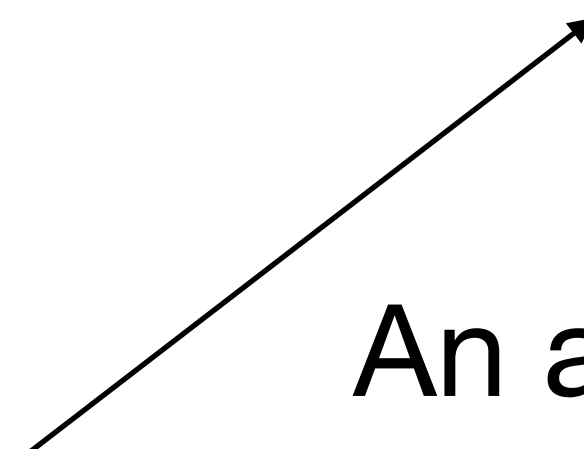
The average percent change



An asset price

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$$dX = (\mu X)dt + \eta X dW$$



An asset price

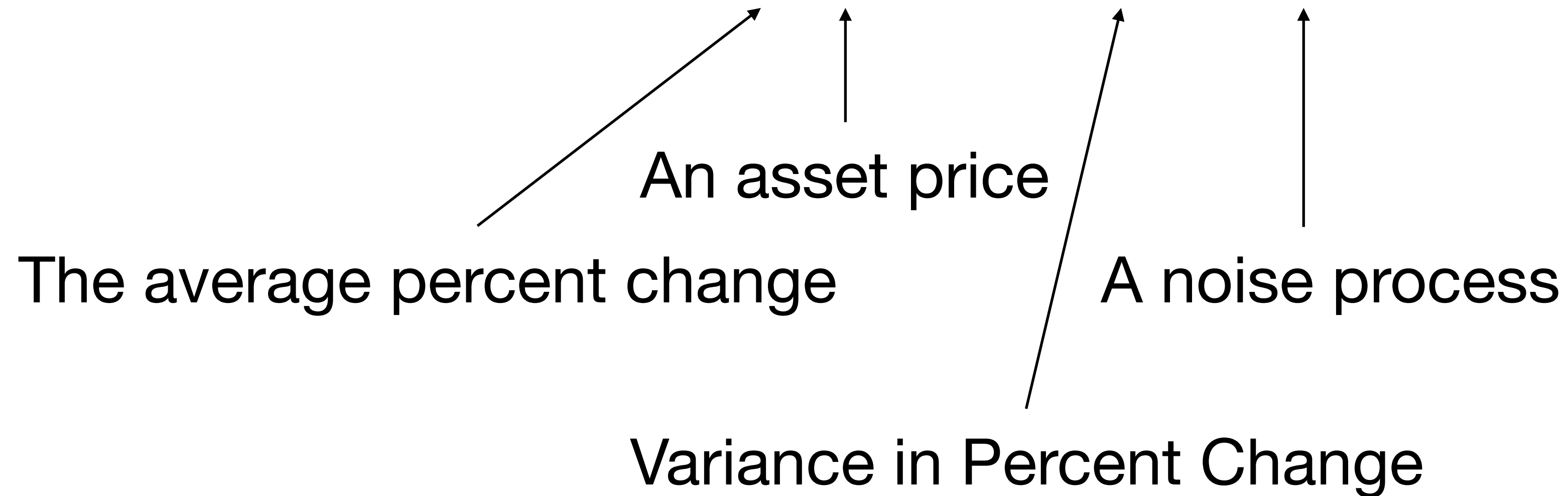


The average percent change

A noise process

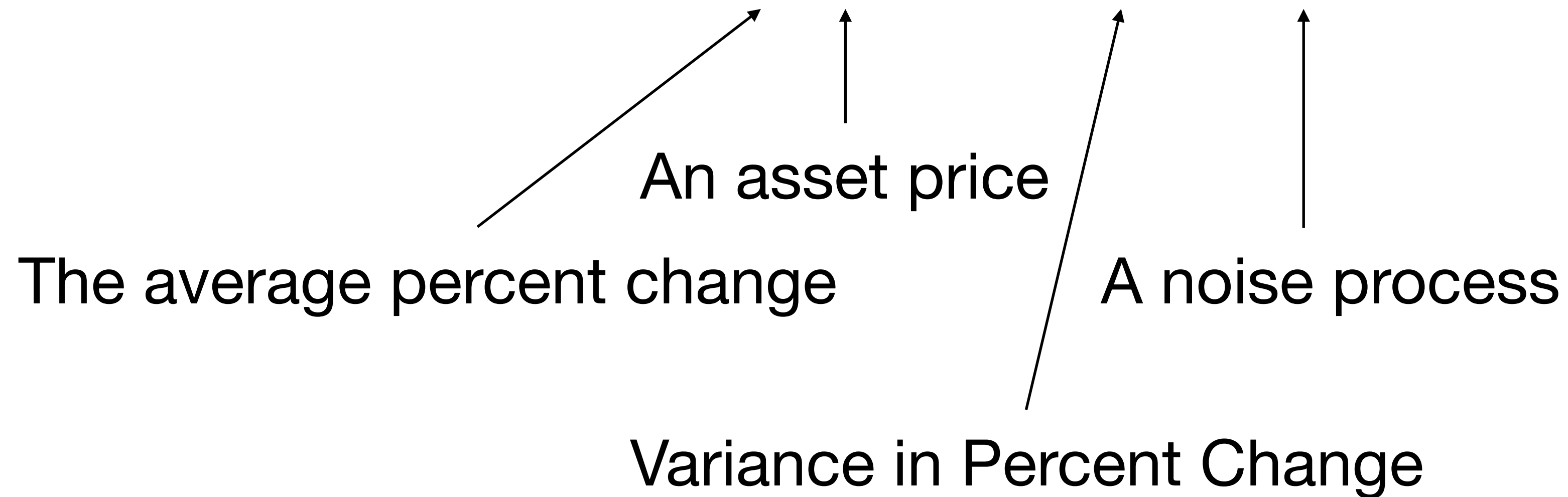
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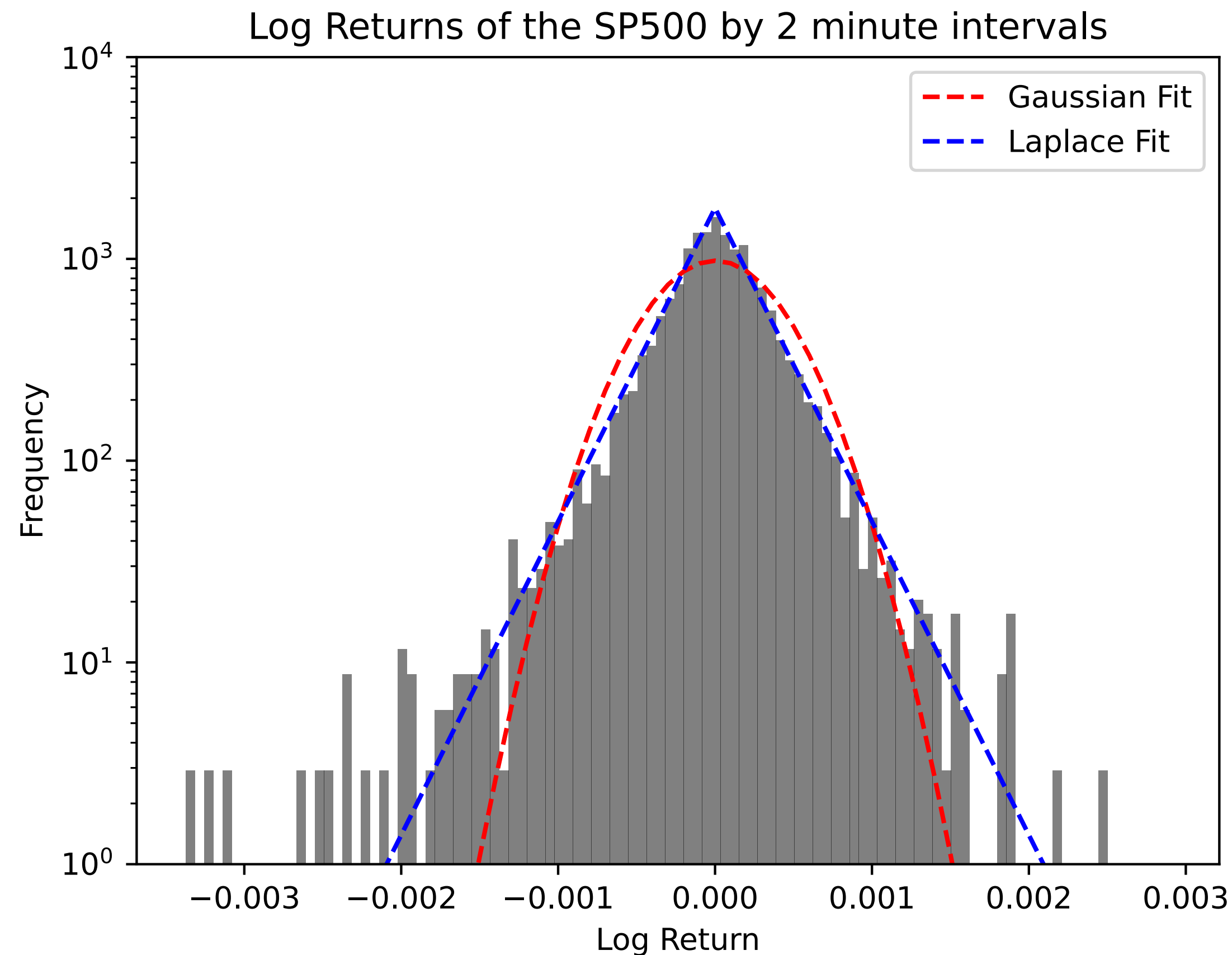


Model predicts that the log of returns is normally distributed

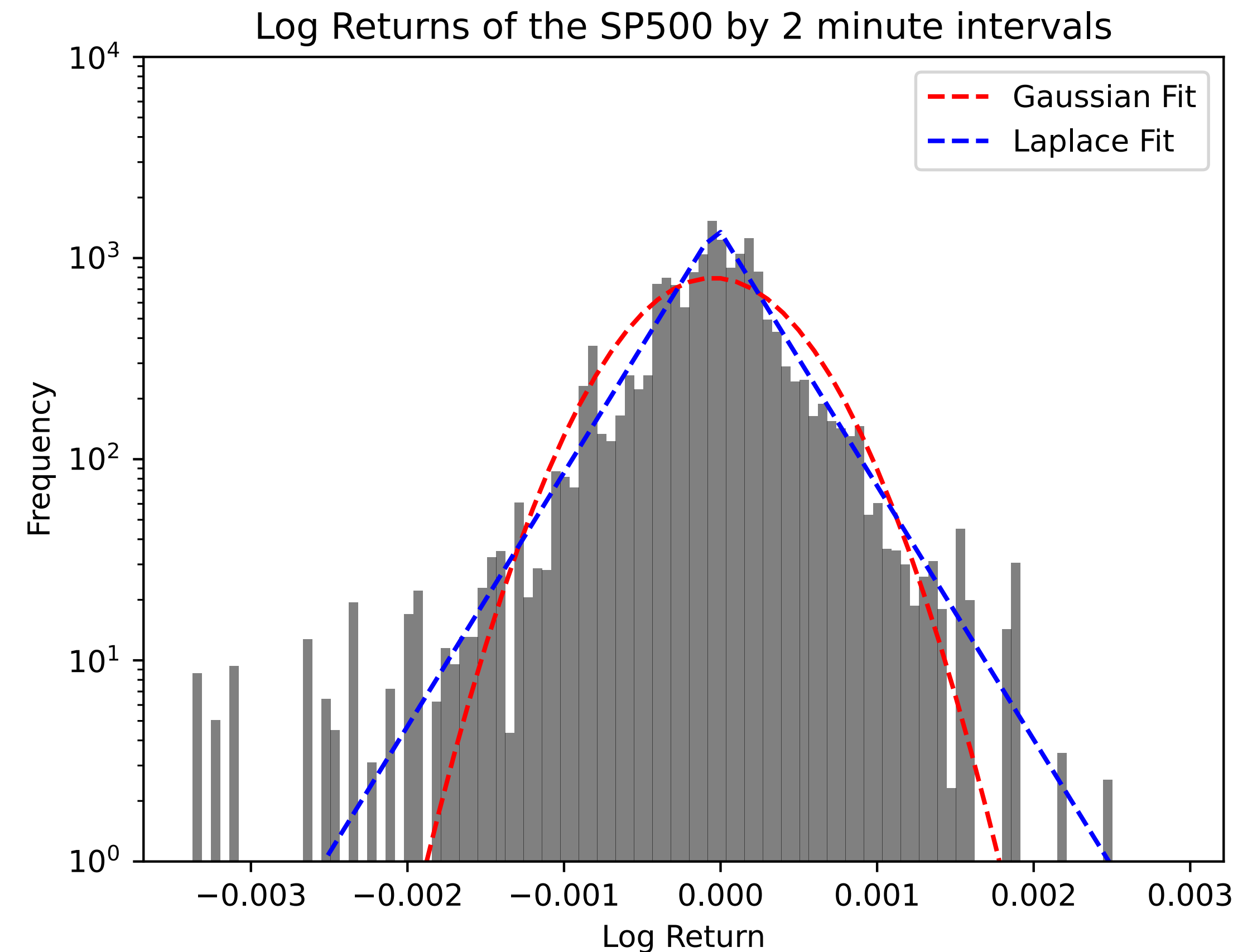
$$\text{Log returns} = \log\left(\frac{X_{t+dt}}{X_t}\right)$$

The observed distribution of log returns

Unweighted by Volume

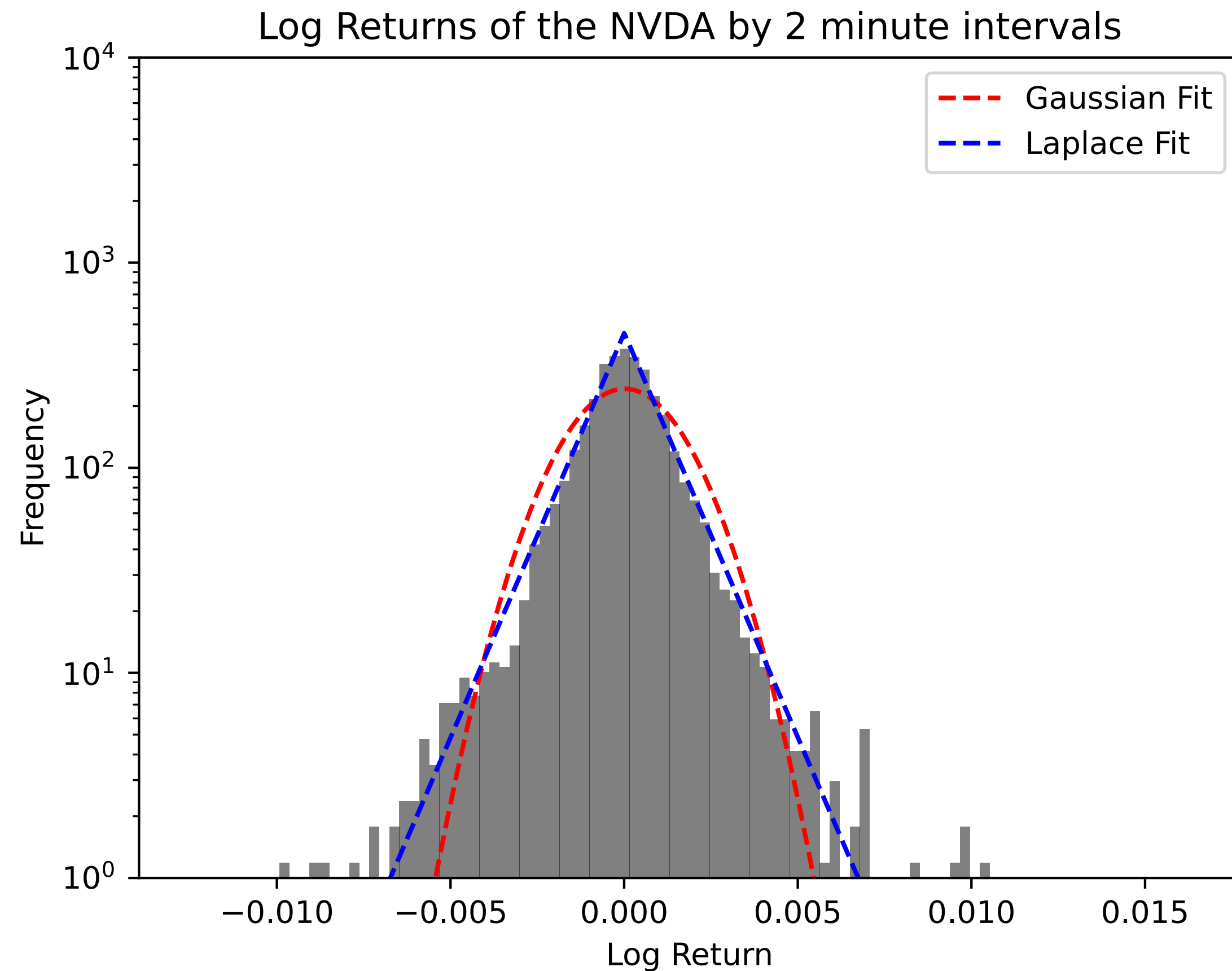


Weighted by Volume

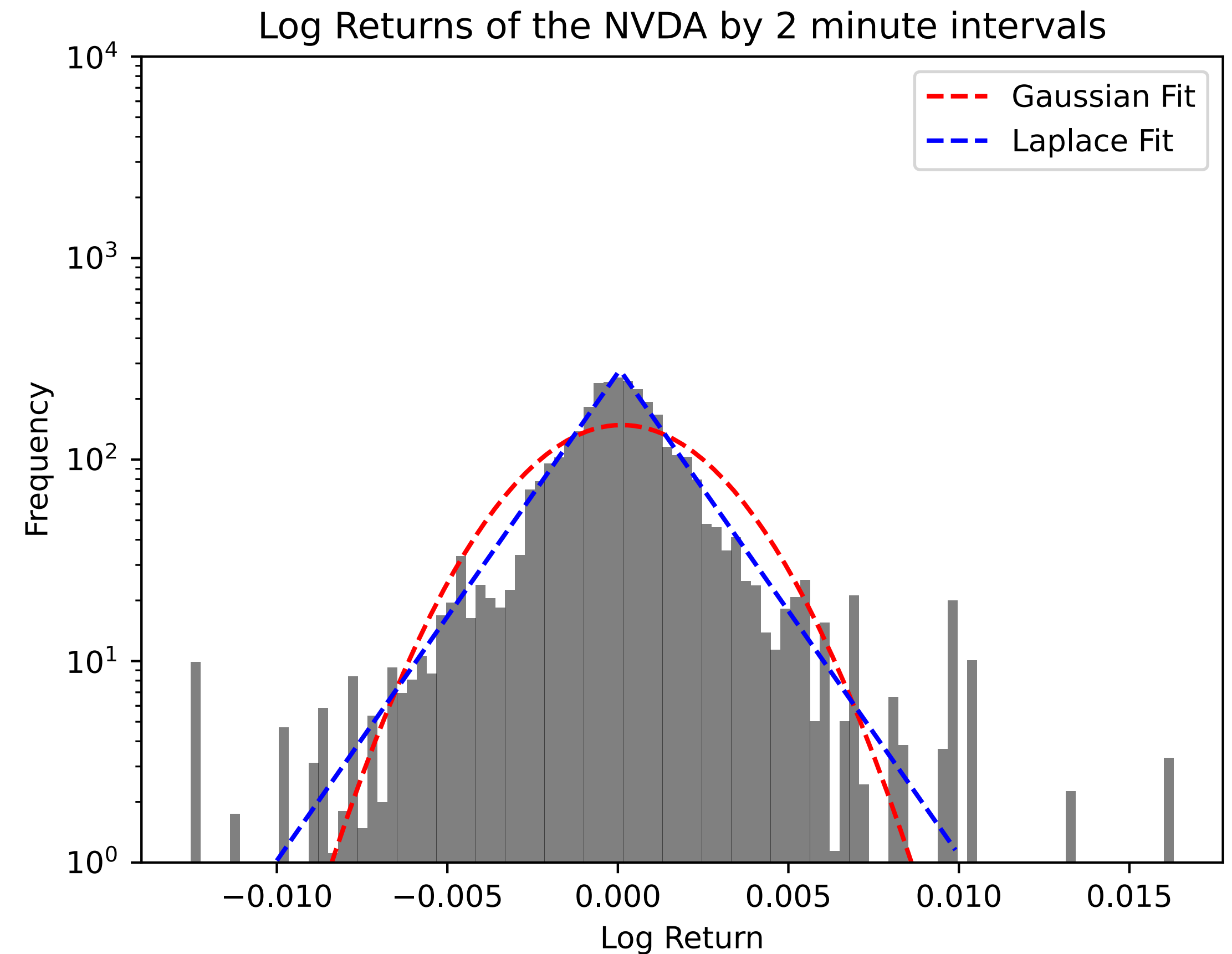


The observed distribution of log returns

Unweighted by Volume



Weighted by Volume



Across Timescales

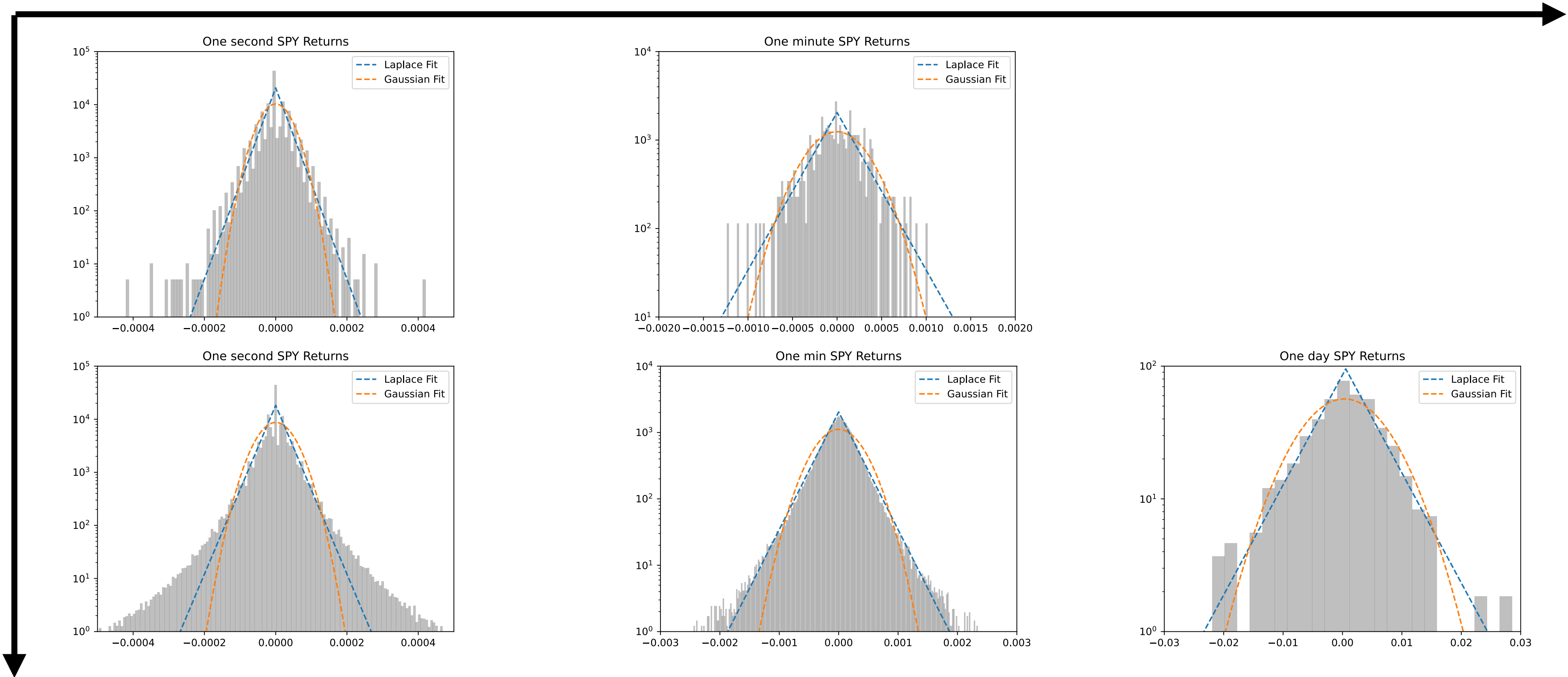
Second

Minute

Day

One Day

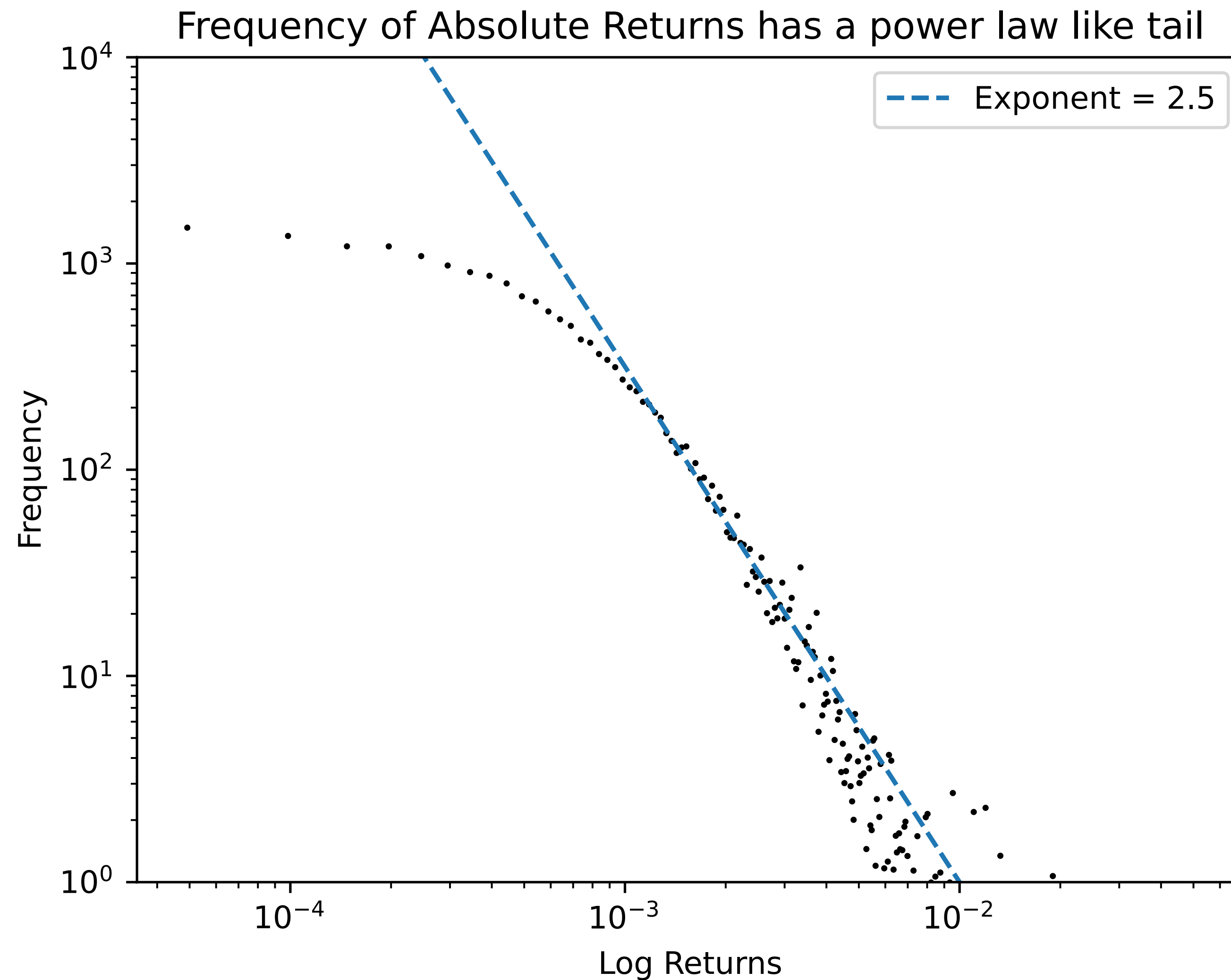
Two Year



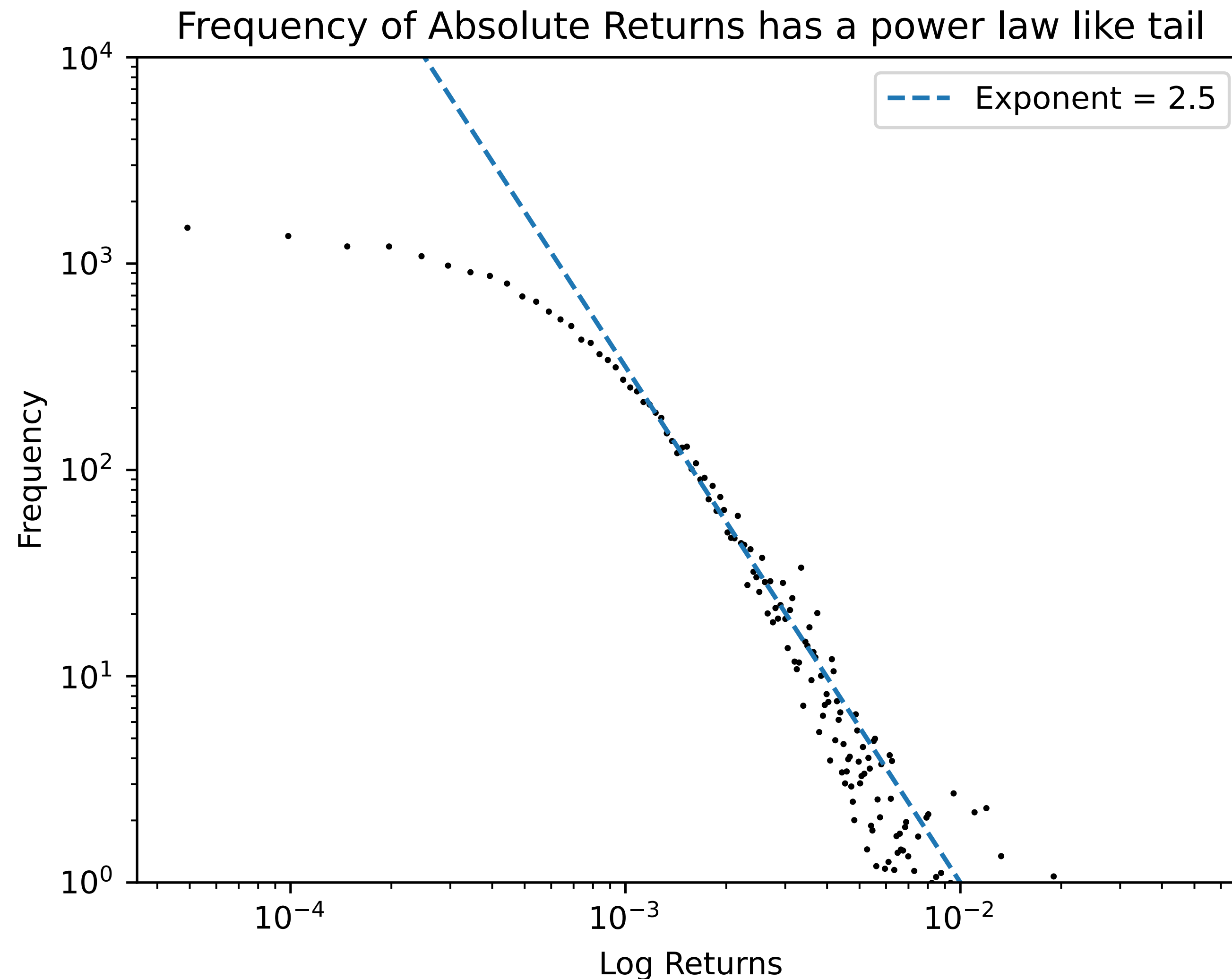
Empirical Observation

- Laplace distributions seem to better describe distribution of log returns than gaussian distributions
- Many potential origins, such as Levy Processes, etc
- Some would go further and consider asymmetric Laplace distributions as the descriptor of log returns

What about power-law tails?



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Power-law tails can result in from mixture of exponential models

Schwab PRL 2014, Aitchinson PLoS CB 2016, Chen Fractals 2019

So how do we predict?

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Using double sided Laplace distributions

$$P(\text{Log Return} = x) = \frac{1}{2b} \exp\left(-\frac{\|x - a\|}{b}\right)$$

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Unbiased estimators

- Location estimator is the median of the dataset
- Scale is the mean absolute deviation from the location

Core hypothesis: Future parameters can be predicted from previous parameters

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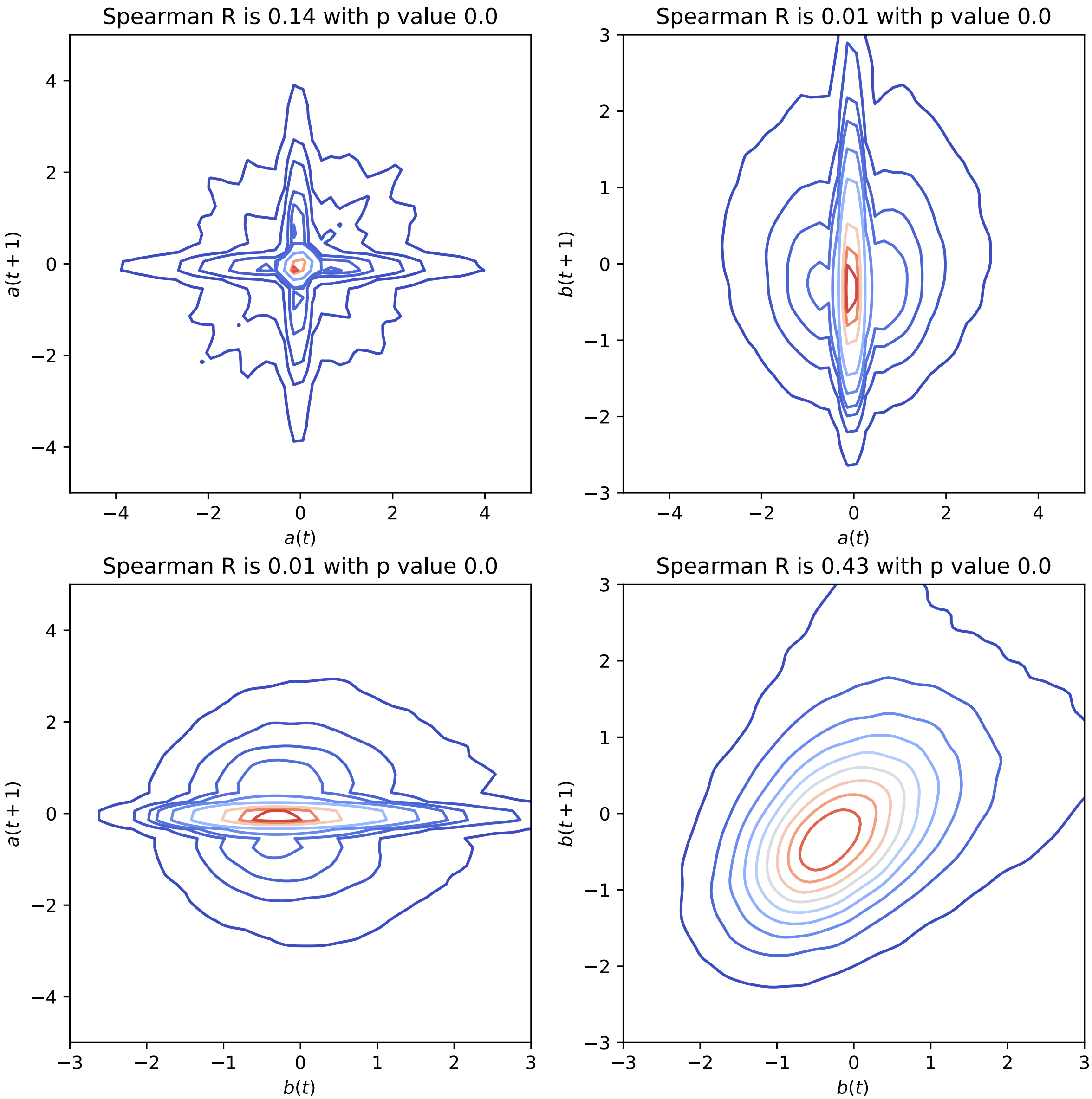
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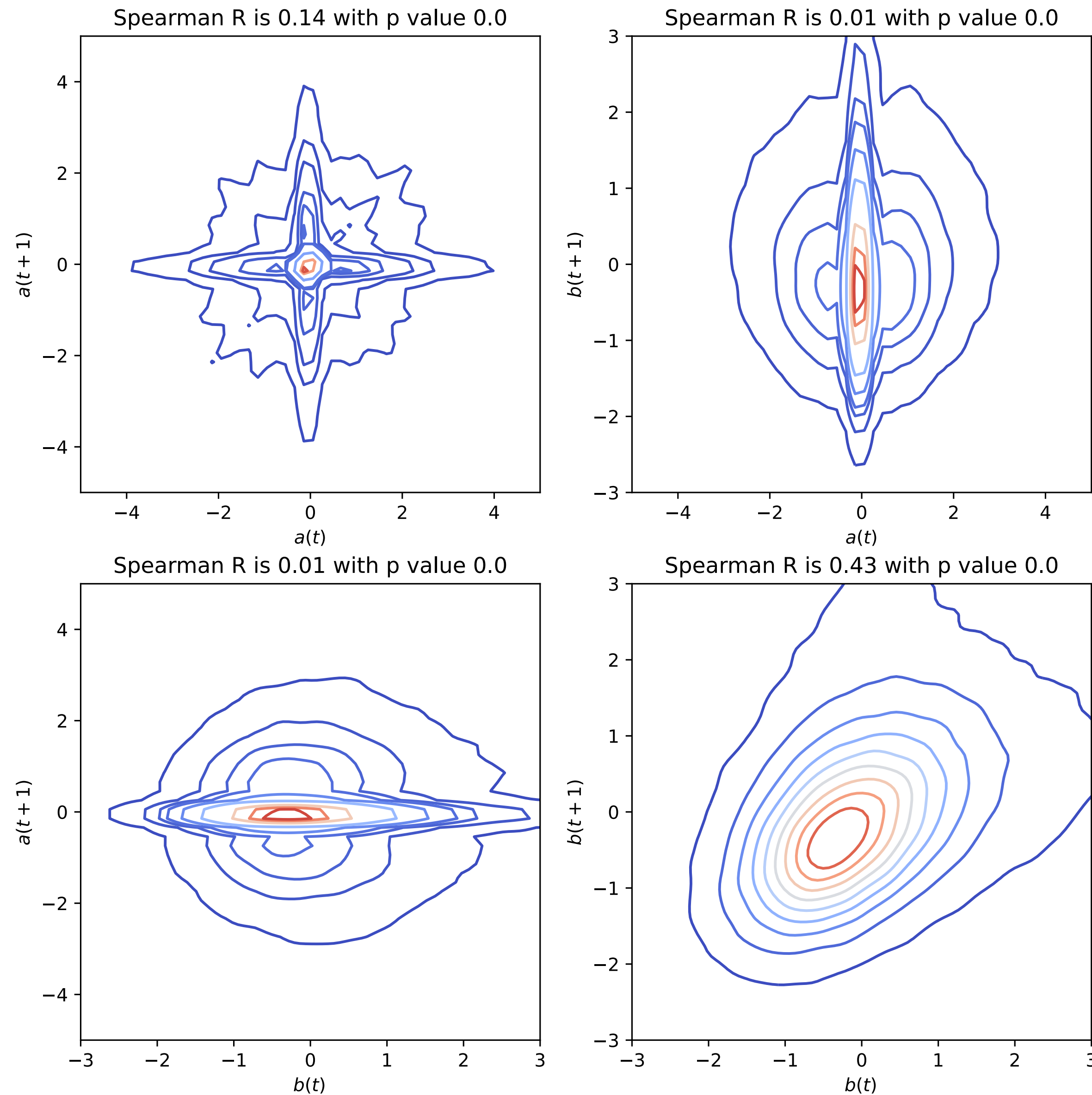
Training data: All SP500 tickers from October 2022 through June 2024.

Testing data: All SP500 tickers from July 2024 to October 2024.

Joint Distribution of Time Parameter and One lagged parameter



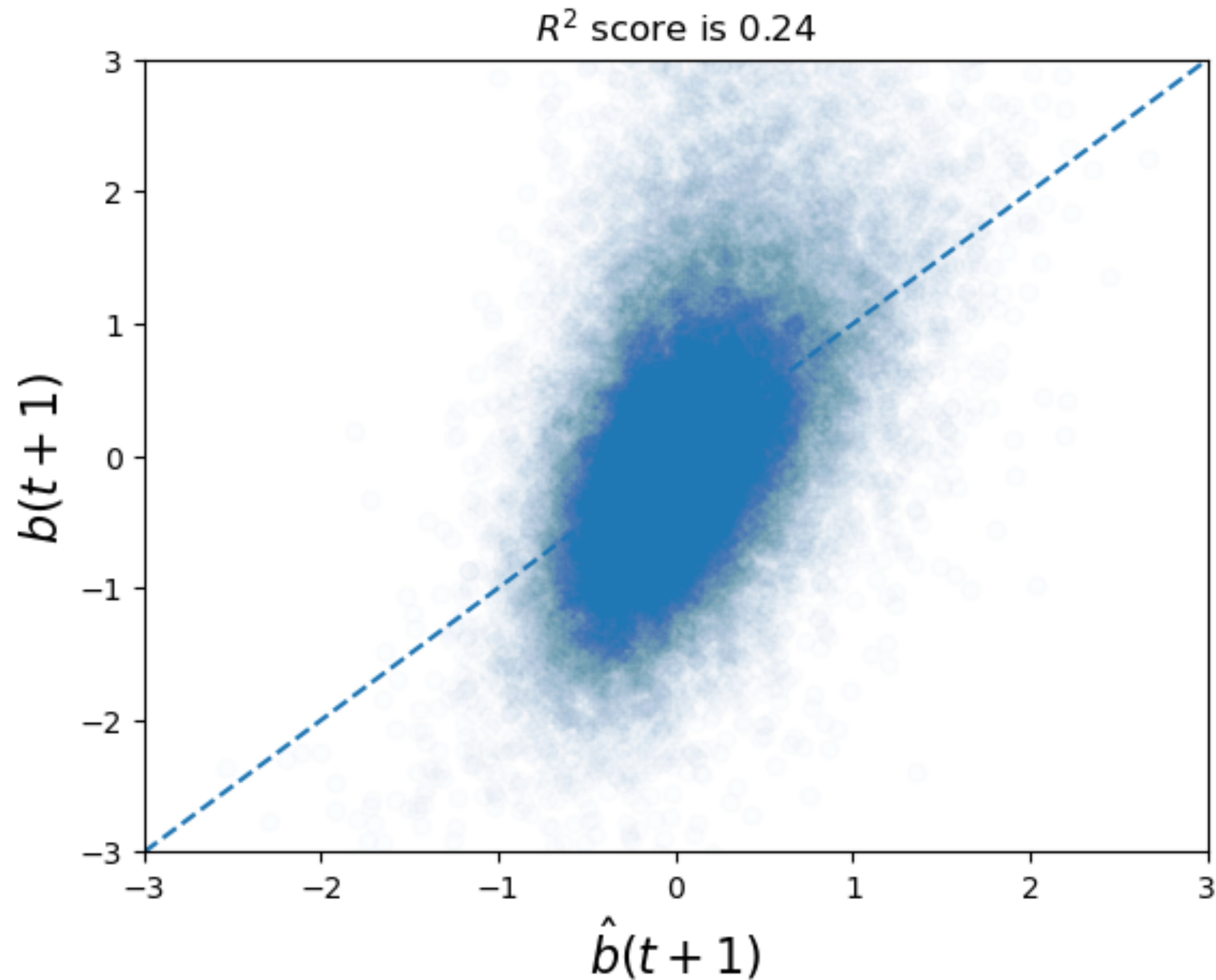
Joint Distribution of Time Parameter and One lagged parameter



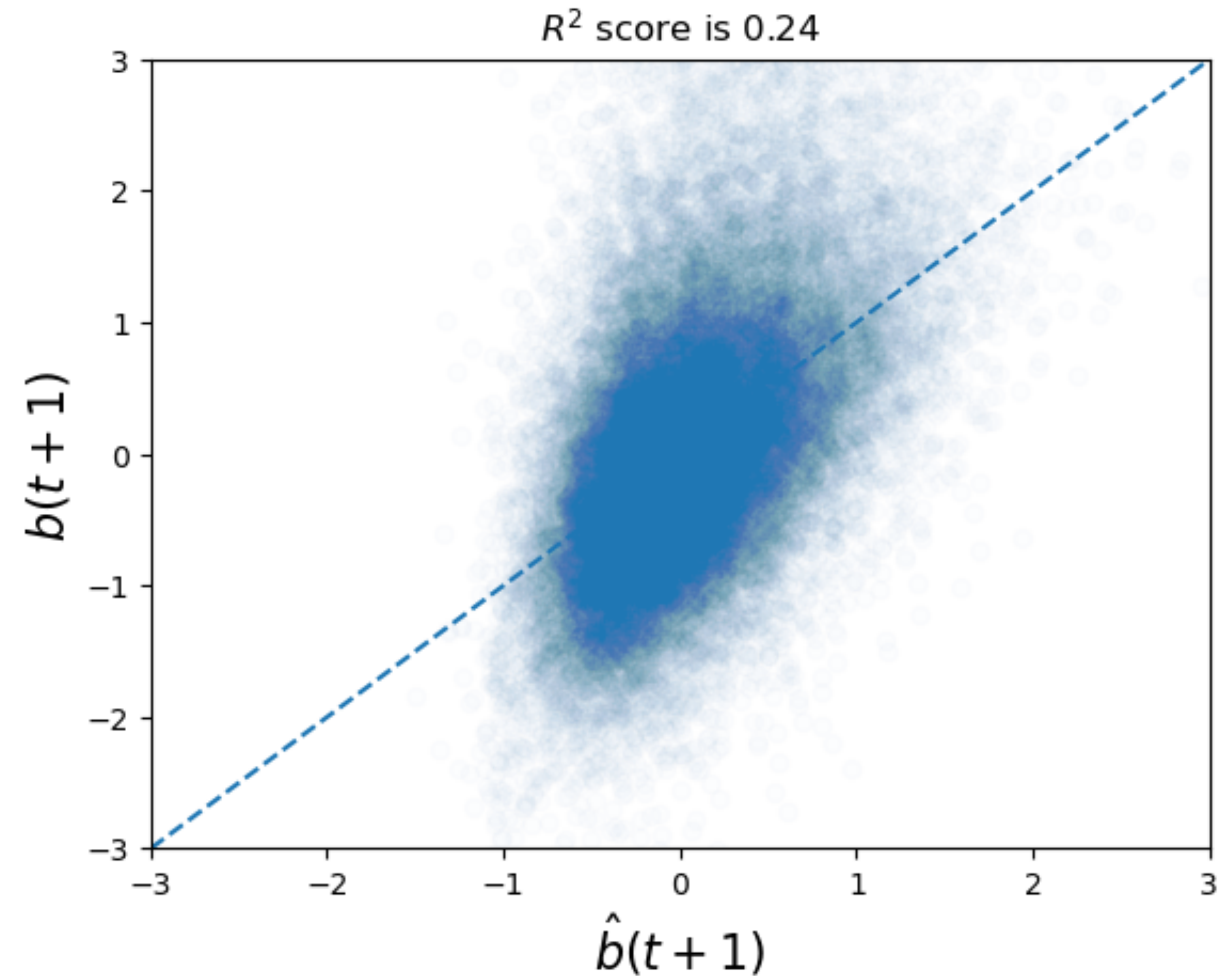
- Scale parameter, b , is autocorrelated
- Location parameter, a , shows weak autocorrelation
- Cross correlation probably too weak to estimate

Building predictor models

Linear Model

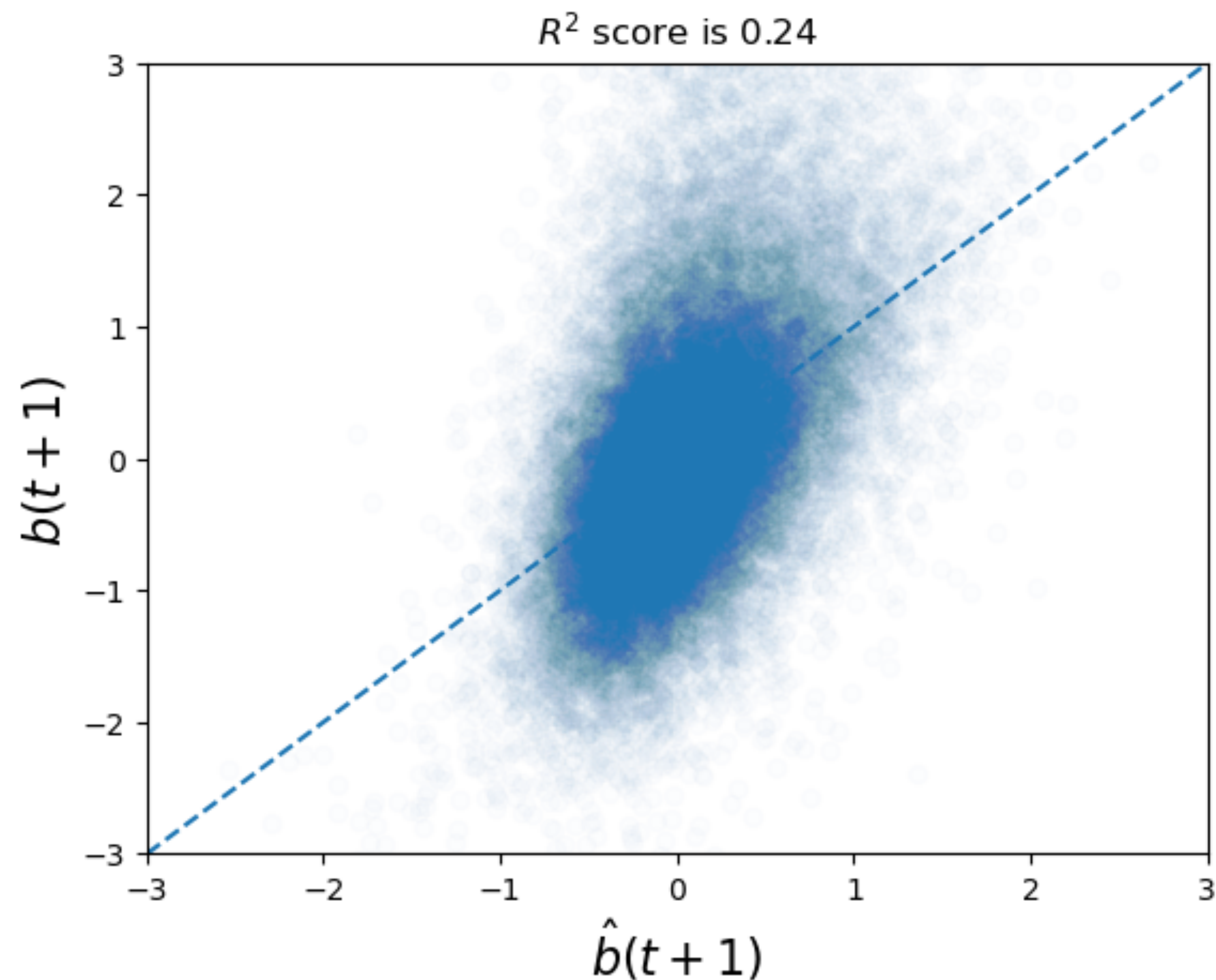


XGBoost Model

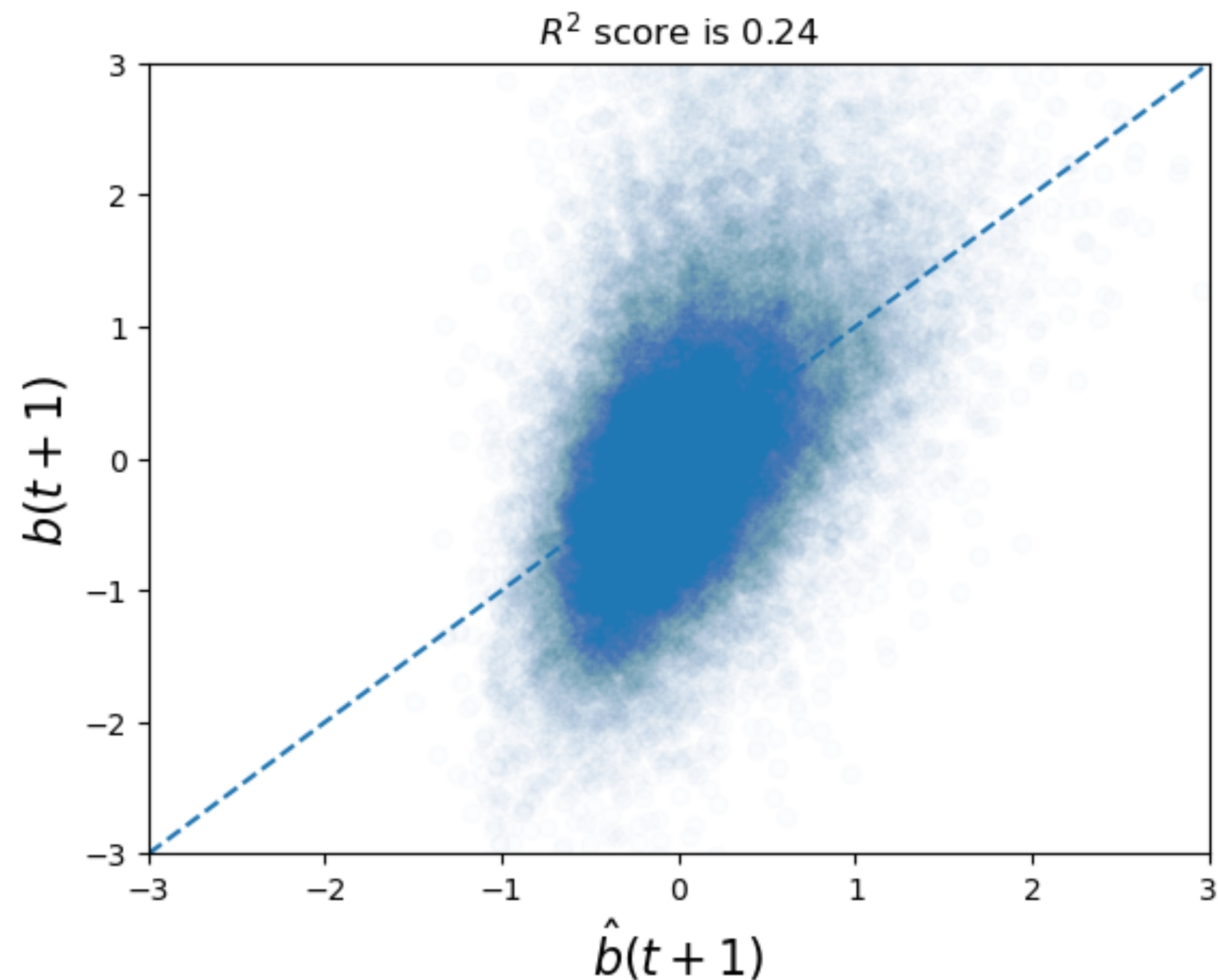


Reasonable models, some room for improvement

Linear Model

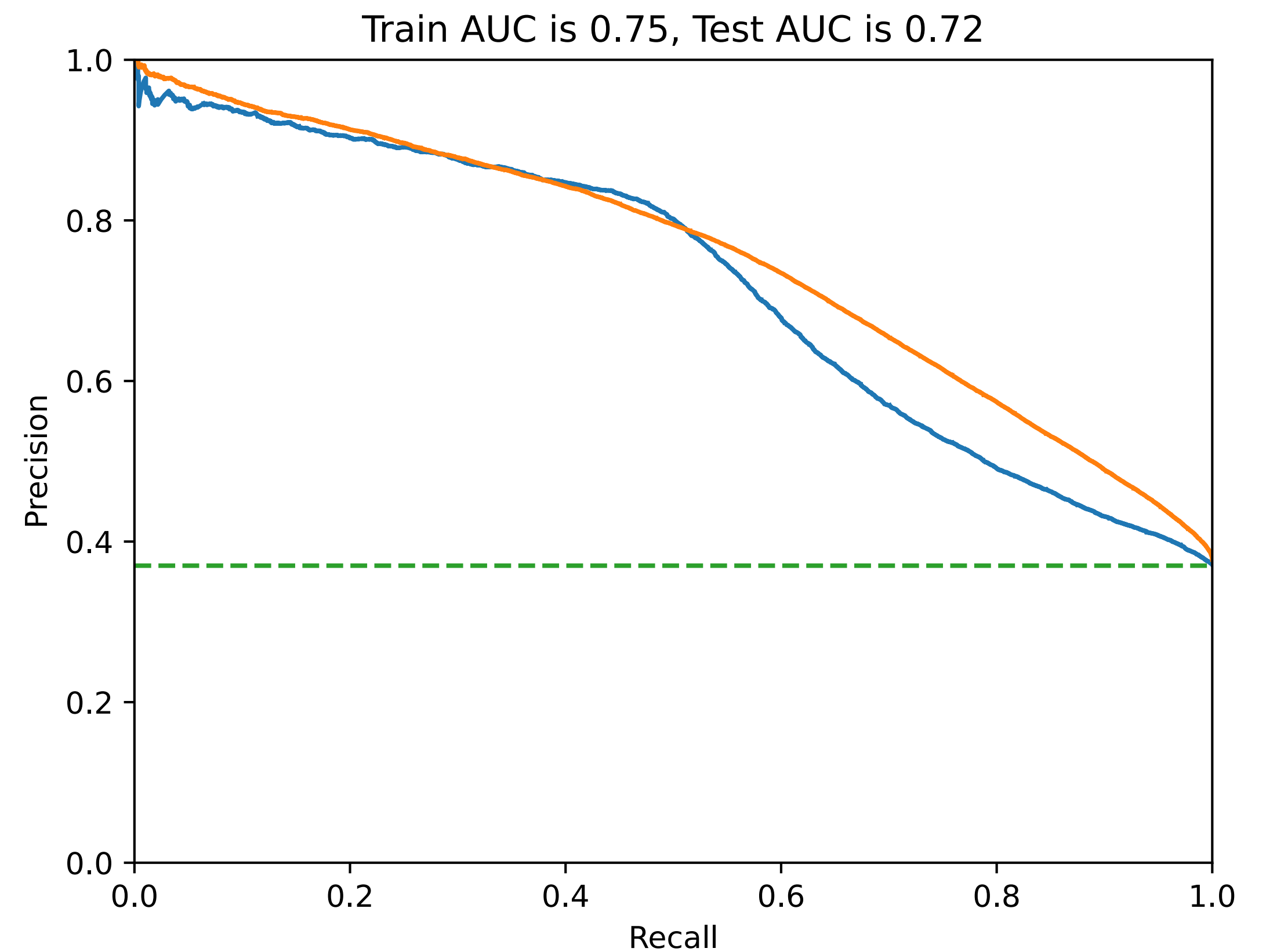
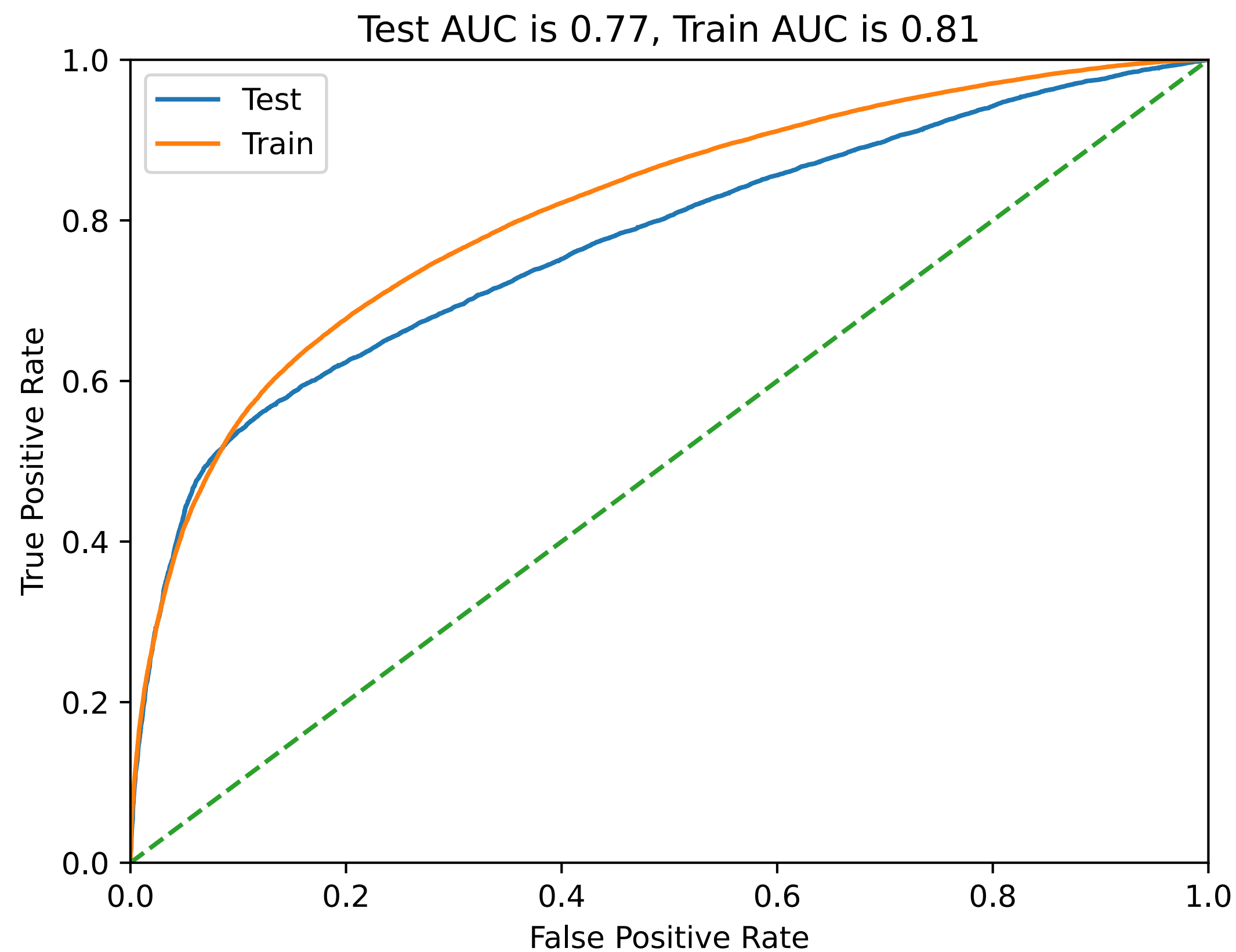


XGBoost Model



Predict if location parameter is greater than 0 after standard scaling the data

XGBoost Classifier



So what can I do?

A trading strategy

$$dX_t = \mu X dt + \eta X dW_t$$

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Replace with Laplacian process

So what can I do?

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Replace with Laplacian process

Integrate forward

So what can I do?

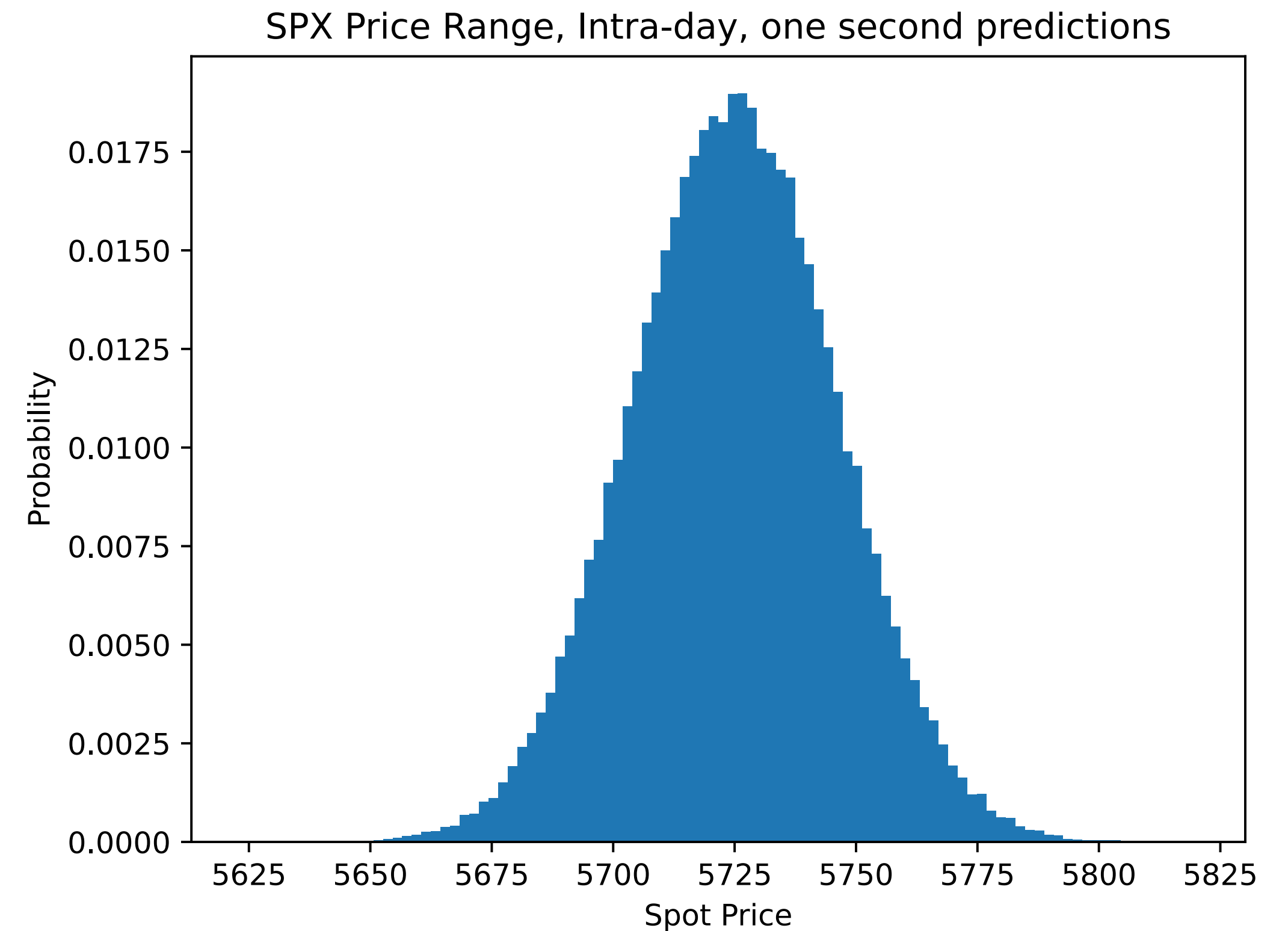
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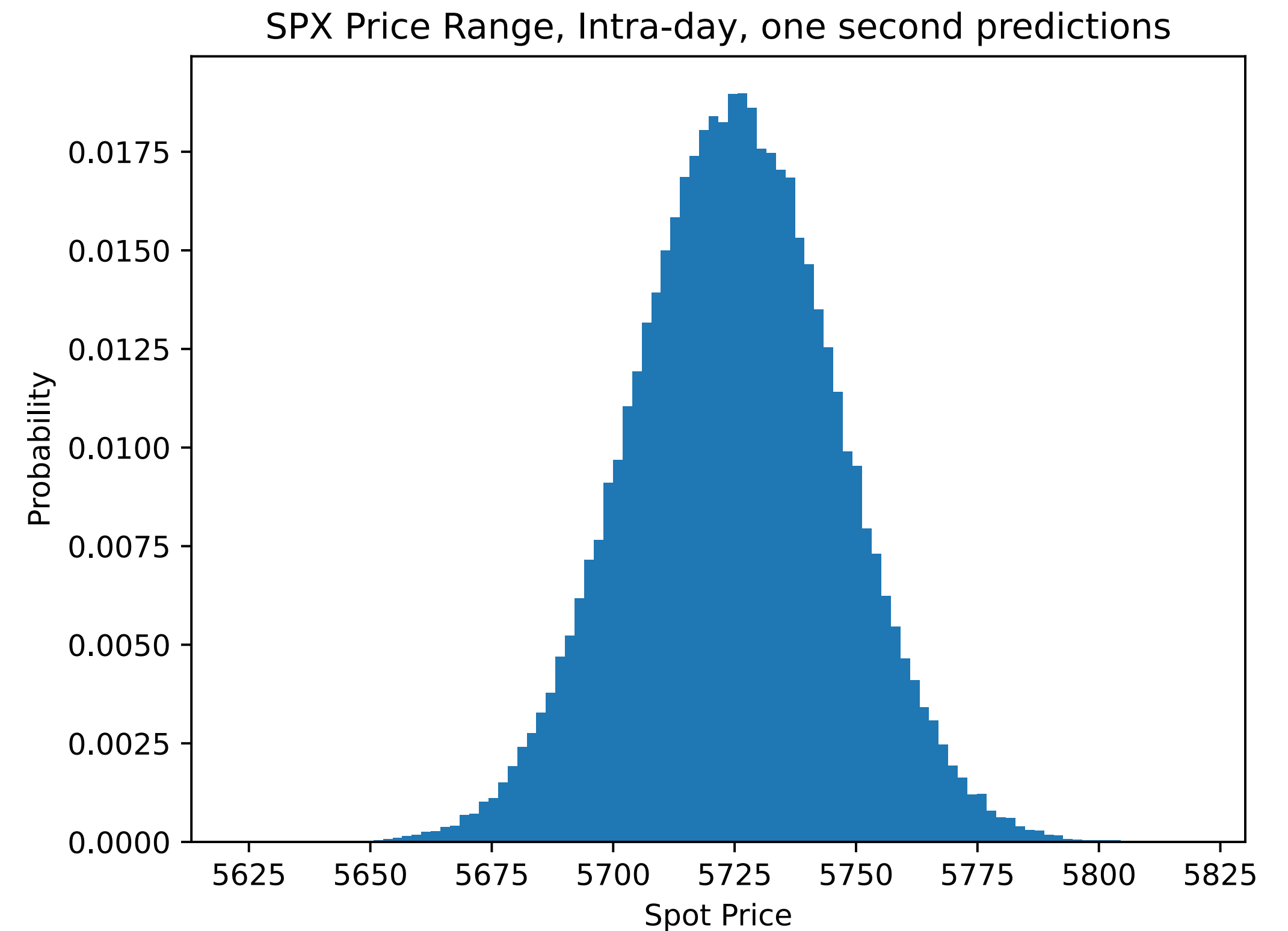
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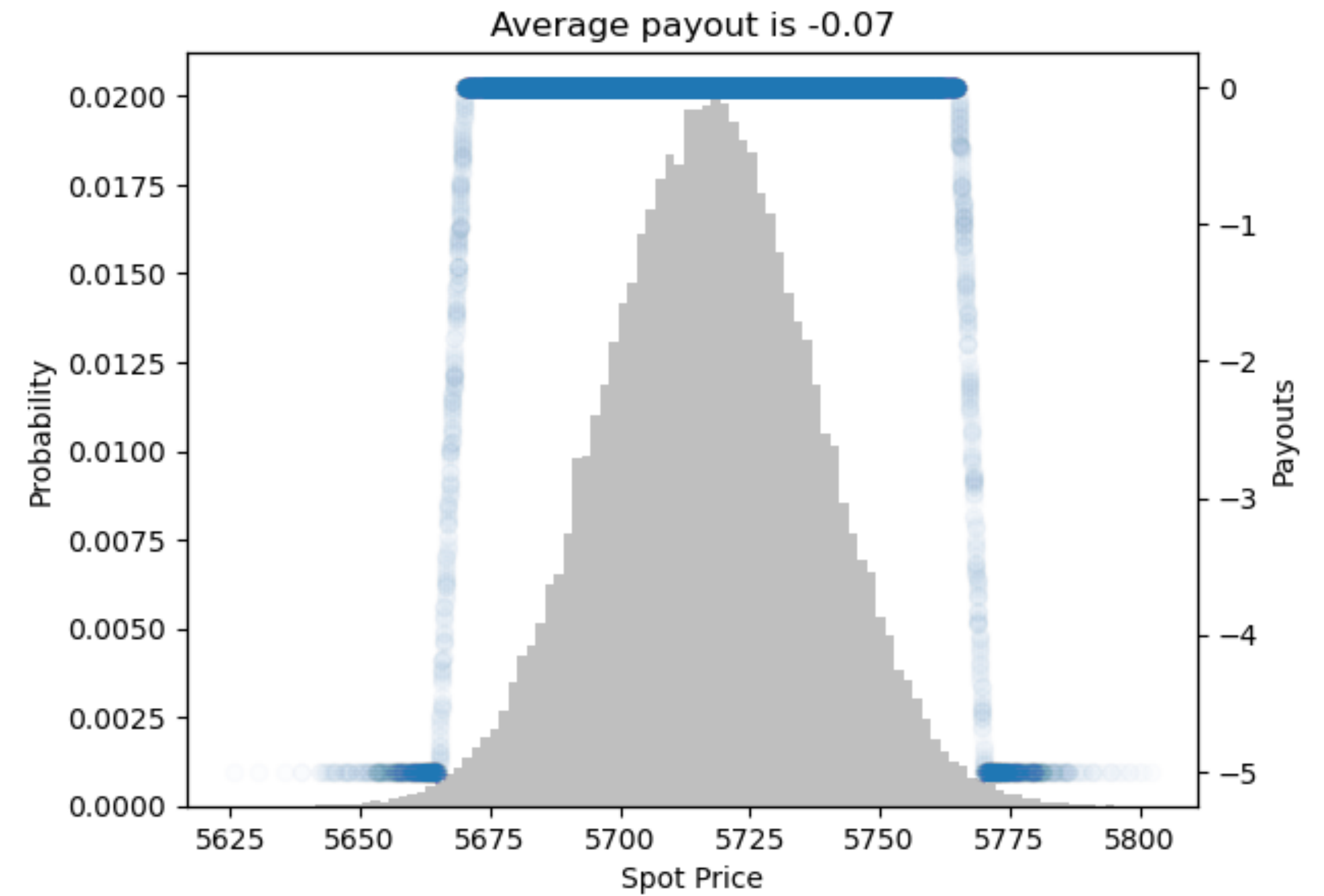
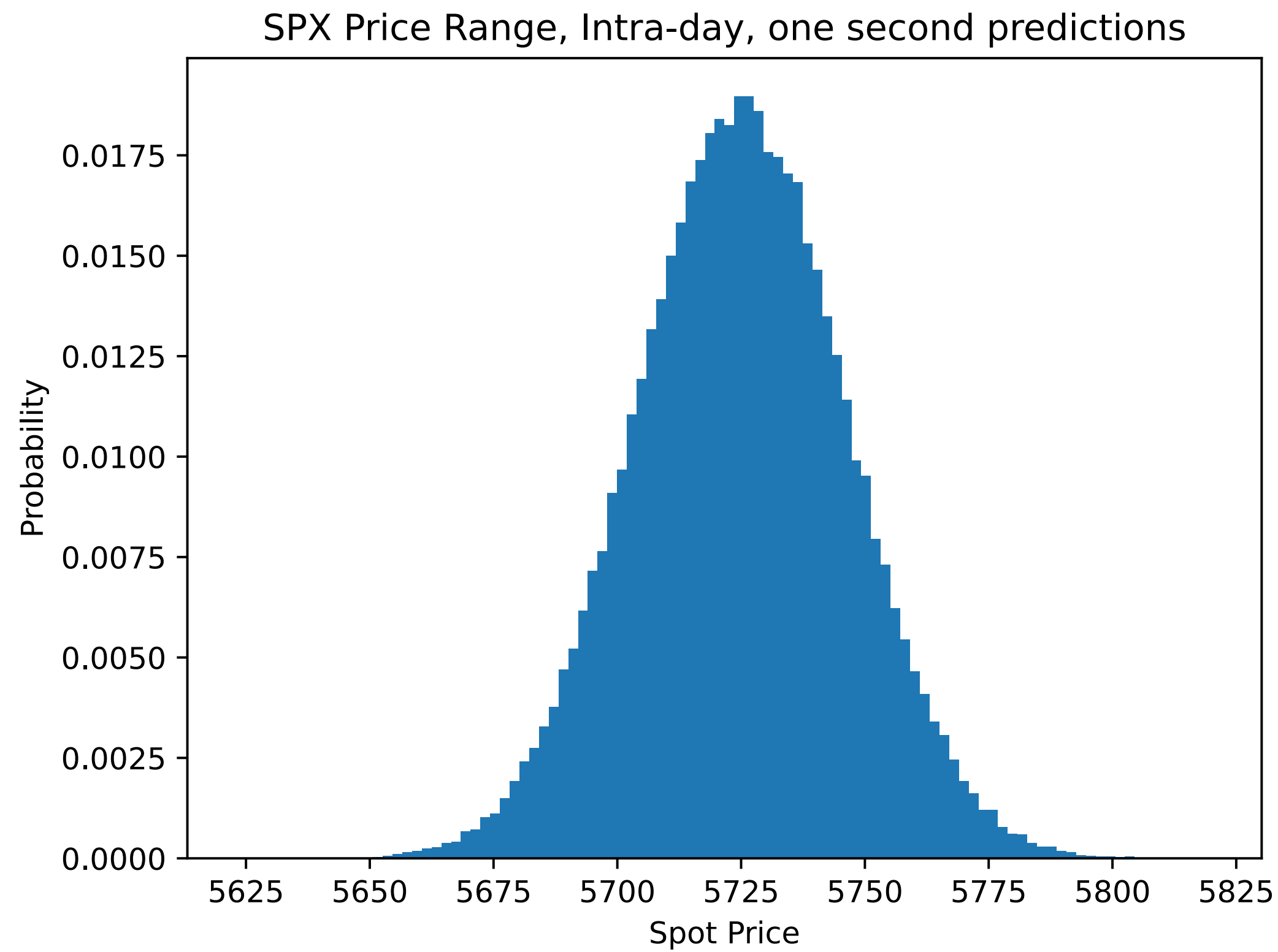
Replace with Laplacian process

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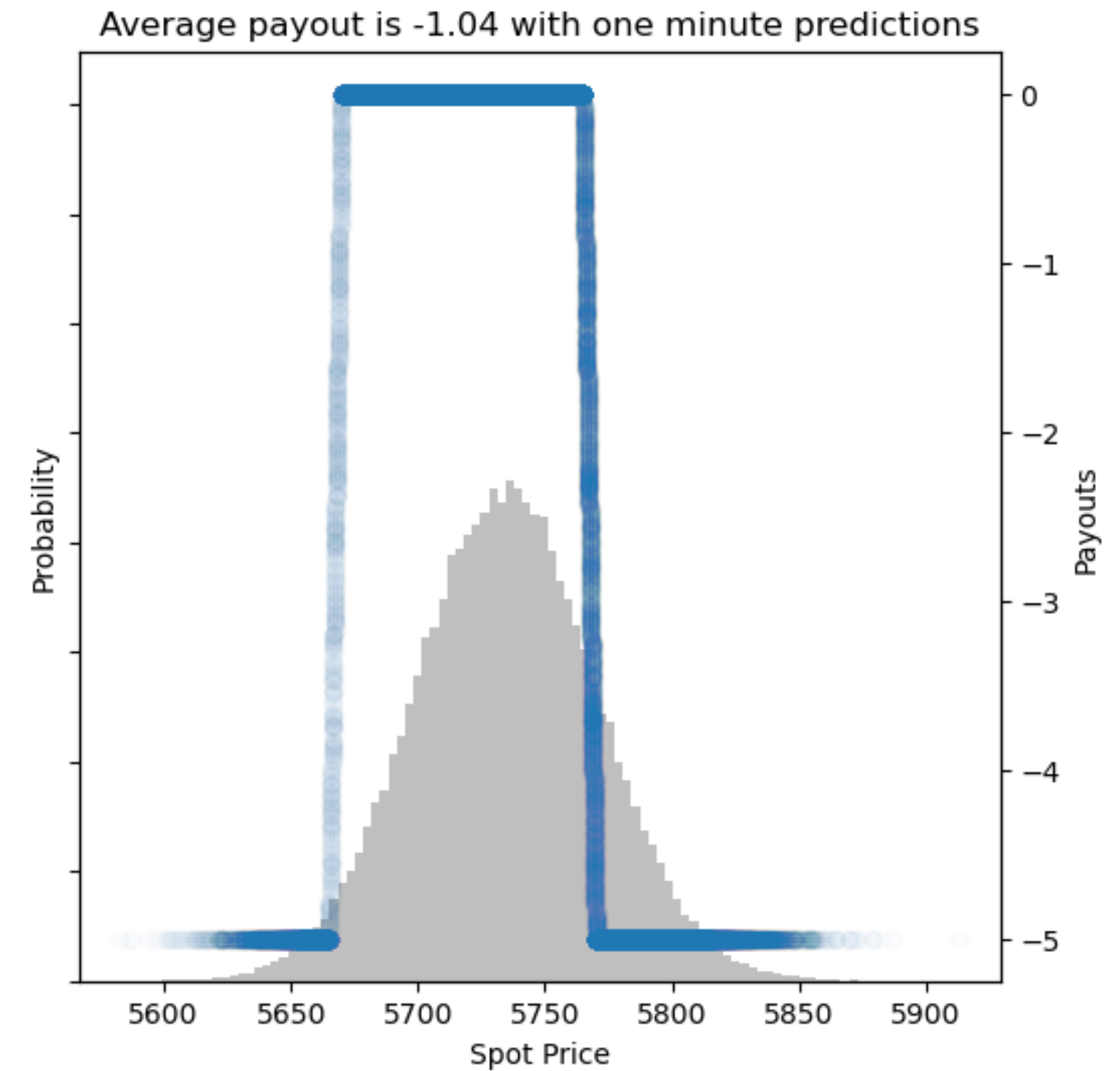
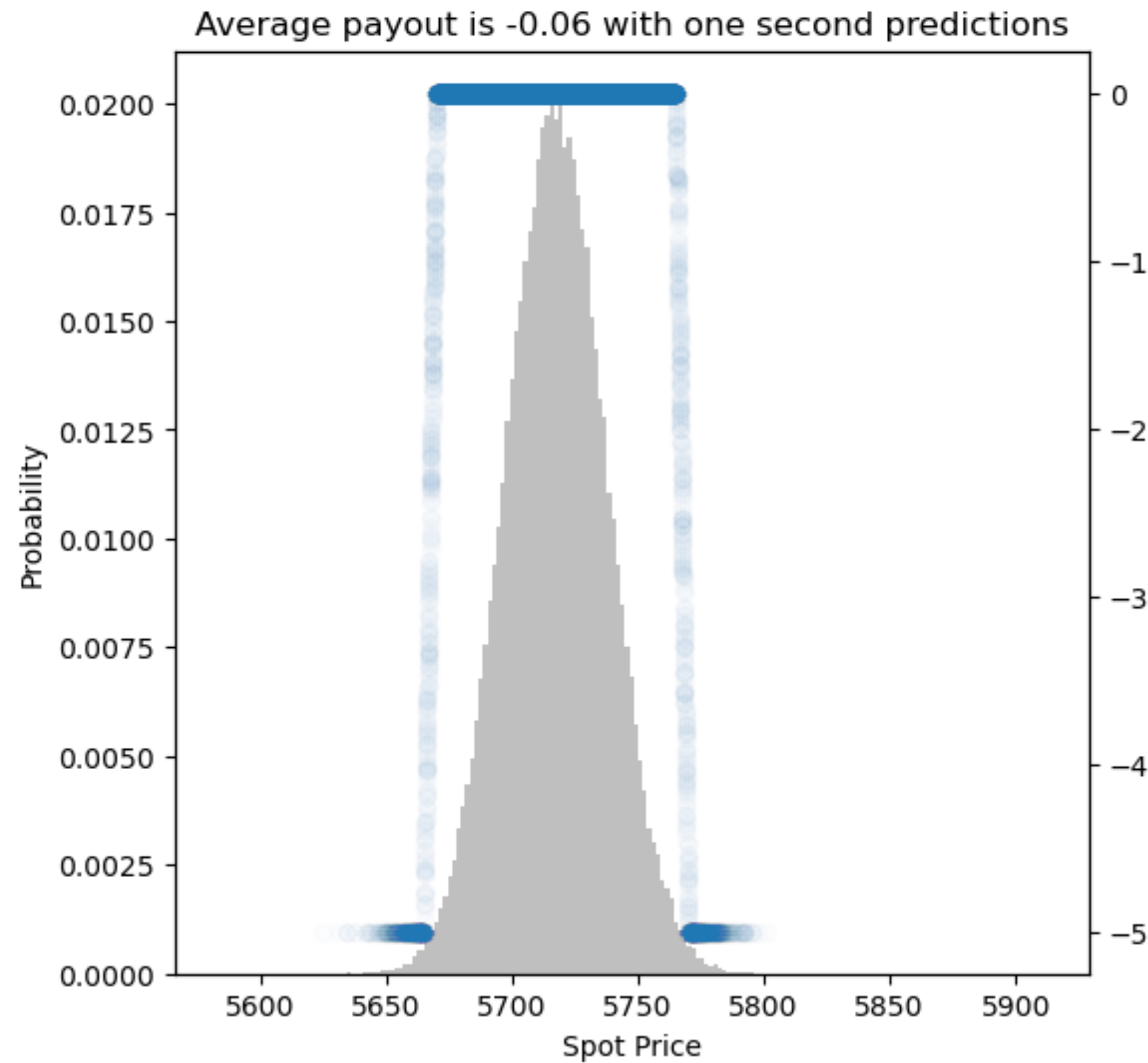


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A trading strategy

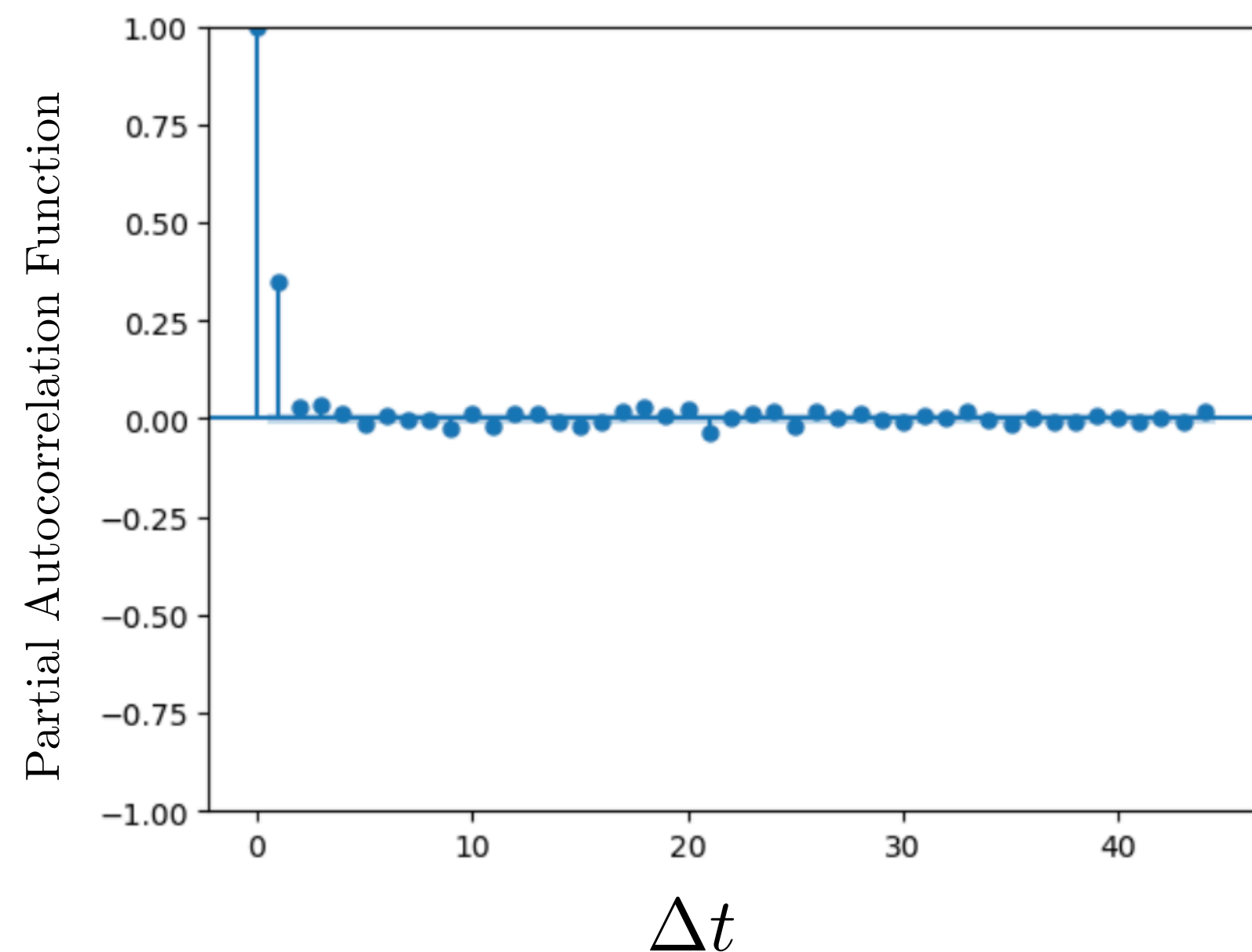


An odd observation

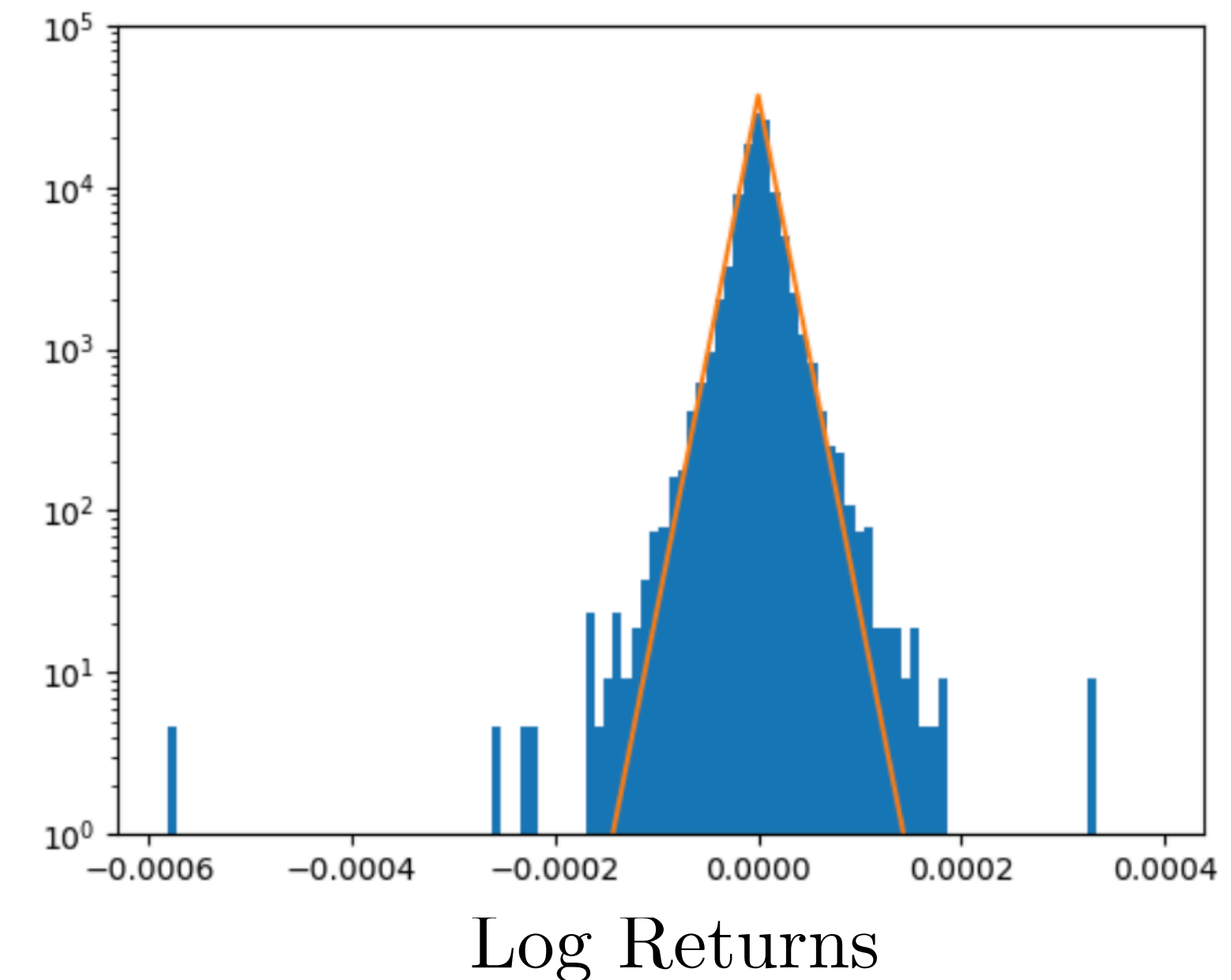


But this fails

- Several assumptions cause this to fail:
 - Independence of each time interval
 - Is the Laplace distribution truly the right distribution?



Frequency



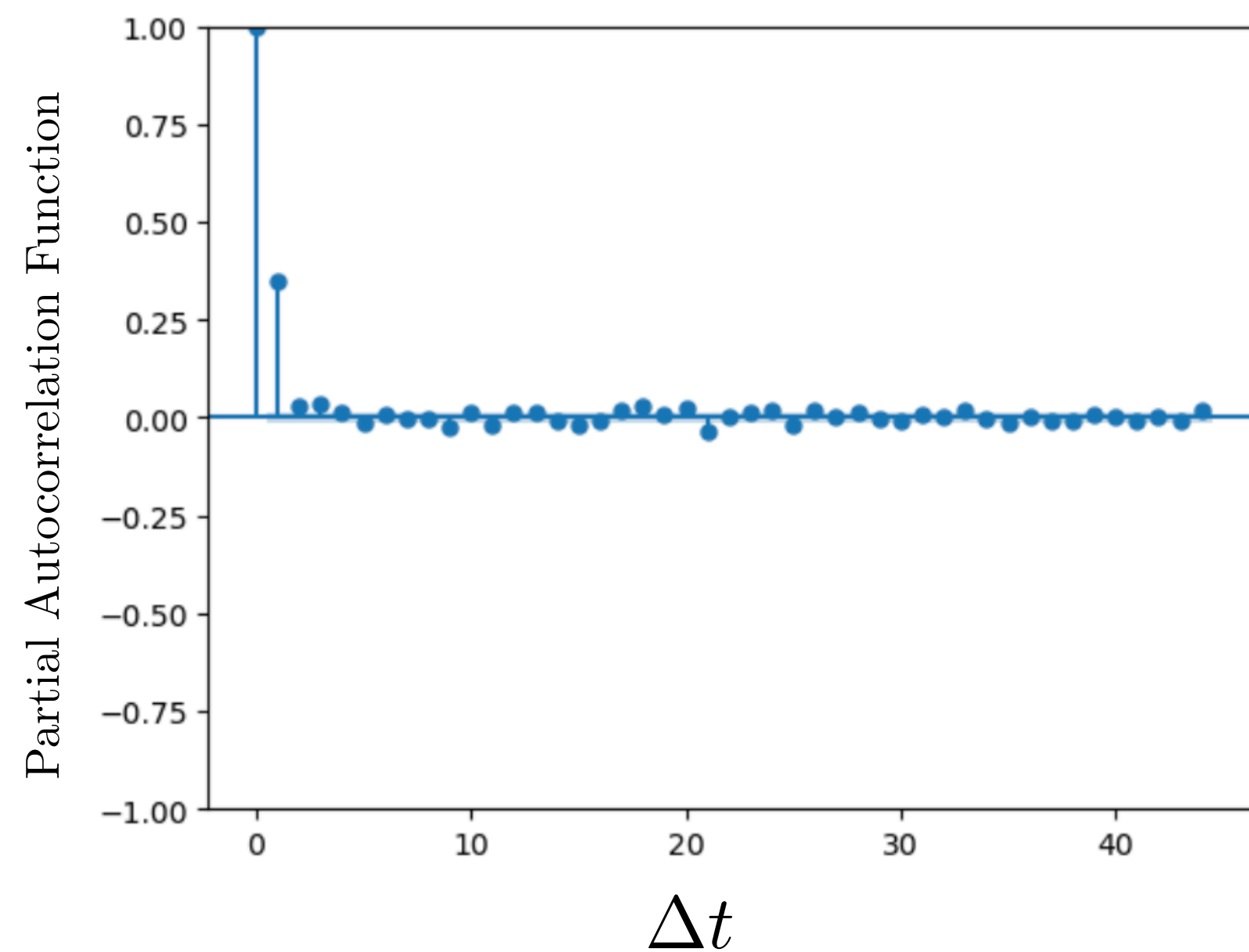
However, it's instructive

- Returning to this formula

$$dX_t = \mu X dt + \eta X dW_t$$

Replace this with a
moment-generating
function

Autocorrelation impacts here



Frequency

